The many dimensions of multimedia communications

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This introductory paper attempts to position the other papers that appear in this edition within the very wide, diffuse and controversial topic of multimedia communications, by relating them to a definition that recognises these differences, particularly in the varying human needs — emotional satisfaction or judgement support — that must be met. Given that no universal solution addresses all of the requirements, it is natural that solutions for entertainment, for example, that use asymmetrical broadband transmission, differ from the symmetrical, switched, more modest bandwidth solutions for business applications such as distributed co-operative working. Architectures and issues for both are discussed.

1. Introduction

Most presentations on the subject of multimedia begin by claiming that the biggest problem lies in the definition of the term, or at least in trying to reach some form of consensus about its scope. For example, some would include television, some would exclude videotelephony, some would do the reverse. Others have different, rational definitions, which are mutually exclusive.

Maybe this is because multimedia, and, even more, multimedia communications, is an entity of many dimensions. For example, it can be described in terms of its technology platforms, their bit rates and connectivity, its vendor markets (communications, computing, consumer electronics), its service attributes (interactive, store-and-forward, volatile, static, etc) and its applications. Later papers in this Journal provide a wide, and therefore sparse, set of points in this multidimension.

It is the purpose of this introduction to plot some of the connecting threads that span the multimedia dimensions, to place the other papers within this connectivity and to identify significant features and issues.

It is the contention of this paper that the principal axis of multimedia communication, the single dimension that best explains its complexity, is its relationship to the human dimension, i.e. how we sense, reason and feel.

2. Definition

Multimedia communications aspires to be a 'worm-hole in space', putting all our perceptions, our understanding and our emotions in direct contact with distant environments, and they with us, but, even in best practice, it is a window of distorting glass that blocks out most of touch and all of smell, and through which light and sound travel slowly and usually preferentially in a particular direction.

With all this in mind, a working definition is offered:

'Multimedia communications, through a range of technologies, including communication over a distance, aspires to provide a rich and immediate environment of image, graphics, sound, text and interaction, which assists in decision-making or emotional involvement.'

This definition recognises the purpose of multimedia communications — it is a support tool to human activity, an activity which can come in different forms and, in particular, emotionally and intellectually.

It suggests that it is unlikely that these needs will be met by a single technology solution.

Also, it adopts the emerging consensus that there are a limited number of media types (text, etc) that are practically available. It is silent regarding whether the presence of any specific media type is a prerequisite for the term 'multimedia' to apply. There is a school of thought that considers image, preferably moving image, to be such a requirement.

It introduces the concept of 'interaction', that is, the ability of the user to have some significant control over the virtual environment to which she or he is exposed. This requirement seems to have crept into common use to distinguish 'true' multimedia from simpler systems such as broadcast television or video cassette. In practice it is a useful distinction, but it must not be used to obscure the fact that broadcast TV provides a benchmark for professionalism of content and quality of transmitted image and sound against which, in the entertainment arena at least, all other multimedia offerings will be judged.

Finally, the definition recognises that distance between the user and the real world poses specific problems which can be alleviated by communications. Multimedia can be provided on a stand-alone basis — for instance, CD-ROM provides a convenient and inexpensive way of accessing still or moving images, users can easily create virtual worlds on desk top computers, and so on. However, in many applications there is a need to update the data quickly and regularly, for people at a distance to work together on the same set of images, or for entertainment services to be distributed to millions of homes. This is where the communications comes in and where solutions which are currently limited in application because of their stand-alone nature may be released into a wider market.

3. Some applications

At this point it may be useful to consider briefly some concrete applications of multimedia communications. The list that follows does not attempt to be exhaustive:

- training and education — the ability to provide access to remote databases and to interact, 'face-to-face', with a tutor located elsewhere,
- distributed work groups — creating a logically unified project team, with common access to multimedia data, e.g. engineering designs and videoconferencing,
- remote experts — providing images from inaccessible locations, e.g. oil rigs, to headquarters experts who can, in return, communicate advice and assembly diagrams,
- information/sales terminals — located in shops, departure lounges, public places, providing information on products or services, perhaps with videotelephony access to a salesperson or customer service,
- interactive television — entertainment, shopping, education and information services based around domestic television and consumer-product interfaces, e.g. hand-held remote controller,
- computer-based information services — as the name implies, built around personal computer access to information services, e.g. Internet.

For a more comprehensive description and extensive bibliography, see Williams and Blair [1].

4. The human dimension

Multimedia communications has been defined as being driven by the requirements of its users. Given that the present state of technology will require trade-offs, it is important to identify and segment the principal elements of these requirements.

4.1 Judgement and emotion

Of course, almost all human activities are a mixture of the intellectual and the emotional, but insofar as they can be separated out in any one activity type, they lead to significantly different requirements. This may be illustrated by a specific example.

Consider home shopping by means of a catalogue — the catalogue can be analysed into three components:

- the illustrations, photographs of the goods, set in attractive surroundings perhaps modelled by flawless models, the prime purpose being to entice you to buy the goods — let us call this the 'seduction' component,
- beside the illustrations are descriptions of the goods, their prices, colours, sizes, etc, and sometimes a diagram giving details of the construction or usage of the product — the prime purpose here is to provide an 'information' component,
- finally, somewhere in the catalogue will be an order form with a postal address or a telephone number, to allow you to place your order — this represents the 'transaction' component.

These components are relevant to paper-based, catalogue shopping and to its more modern equivalent, 'electronic shopping'. Using the terminology of the latter, these three 'requirement' components are mapped, in Fig 1, on to the 'solution' components that are required in order to deliver them:

- multimedia,
- databases,
- communications.

It should be noticed that the multimedia component has a major contribution to make to the seduction element and also a significant one to the information part, reinforcing the implication of our definition that multimedia affects emotion, that is, 'seduction', and also assists decision-making, the 'information' part. The multimedia component in the seduction element aspires to be exciting, vivid, movie quality, coloured, audibly rich; the multimedia component in the information domain should be clear, informative,

requirement / solution	seduction	information	transaction
multimedia	exciting, high quality moving images and sound, animated graphics face-to-face videophony	graphics, stills, spoken instructions	
databases		hold content and results of interactions but do not directly interact with people	
communications	access to up to the minute/live feeds, remote assistants, counsellors	access to remote information, updates, remote experts	order placing, payment taking, delivery of electronic goods, negotiations with remote sales points

Fig 1 Mapping of solutions against requirements.

perhaps an animation, but not necessarily moving video. There is almost certainly an abundance of text and/or line diagrams. (This is not really a new discovery — there is a well-known adage in publishing regarding the use of colours: 'Two to tell and four to sell'.)

In general, the required richness of media for decision-making is less than that for emotional involvement.

It must not be assumed that moving picture quality is the single dimension of overall emotional stimulation. People often need fast, two-way interaction to achieve emotional satisfaction, e.g. in achieving empathy. The one-way transmission of images, from machine to human does not fully meet this need, nor does a message-based, person-to-person video link. Sometimes conversational voice or videotelephony is required.

Furthermore, an informal body of evidence is emerging, in conversations with advertising and media bodies, that the quality of audio, in terms of its content and its recording and reproduction, is a major element in providing emotional satisfaction.

The contribution of audio to the multimedia environment is an under-researched, or at least under reported, area of study.

4.2 *The asymmetry of content*

Even if media quality, in terms of clarity of image and sound were a necessary requirement for a satisfactory emotional experience, they would not, on their own, be sufficient; to paraphrase slightly a technical director of Sega, the computer games company:

'The games platform is only an enabler; apart from price, three further things are required: content, content, content.' [2]

The creation of high-quality content, either for games, TV or any other entertainment service intended to capture our emotions, requires rare talent and considerable expense. This applies both to the performers in multimedia entertainment and to the technical and design people involved in putting together the structure that surrounds the performance, whether it be a movie, home shopping, banking or other service. The paper by Smith and Jones [3] describes a typical service creation environment for the generation of such offerings.

Specifically, the paper distinguishes 'service design', i.e. the generic infrastructure, from 'application creation', i.e. the design and creation process which is undergone anew for each 'shop', 'bank', 'cinema on demand' offering. Service design encompasses the commercial issues of the scope and scale of the service, including how it is to be branded and what ordering, payment and other transactions are permitted, technical issues such as platform type and media capability, and user interface issues such as generic rules for navigation and screen layouts.

Completion of the service design results in the creation of a number of documents which are used in the application creation stage. This addresses the construction of specific offerings from individual retailers. Typically, it makes extensive use of storyboards and workshops involving creative designers and people who understand the capabilities and limitations of the platform.

This process can be described as: ‘Trying to produce a fun offering, on time, on budget’. More graphically, it can be described as: ‘Getting the Bolshoi Ballet (the creative artists) and the Red Army (the database specialists and systems programmers) to work together under control of the KGB (the management)’!

It follows directly from the difficulty and expense of creating absorbing multimedia material that there will be a decided tendency for there to be more receivers of quality content than there are transmitters.

As a corollary of this, it follows that content-based multimedia services will be highly ‘asymmetric’ — there will be a requirement for large data flows in one direction and not in the other. Also, it becomes cost effective to place the burden of cost on the relatively few production equipments and aim for low-cost receivers, as will be seen when video on demand is discussed. Related to this asymmetry, it also follows that the content will be created and stored off-line for access by the user at the most suitable time.

These statements naturally lead us to a store-and-retrieve architecture, rather than a fully bi-directional, low end-to-end delay service. Thus, there will be a tendency to support standards such as MPEG [4, 5] media encoding, with its high complexity of coding, low complexity decoding, and its relatively long buffering requirements, rather than the H.261 standard for videophony.

In order to capitalise on the reuse of expensive content, its creators will attempt to sell their creations across a number of platforms — CD, TV, etc. Two consequences follow directly from this.

- Firstly, a standards issue — developments in the computing, consumer electronics and entertainment industries can be expected in order to seek standardisation at the video and audio encoding level, however reluctant they are to lose differentiation [6]. The Betamax/VHS wars haunt the industry and it is unlikely that domestic consumers will tolerate such a thing again. It will be interesting to see whether the current disagreement over compact disk standards is terminated quickly before significant competing products go on sale [7].
- Secondly, there will be a need for platform-independent ‘service creation tools’, to make it easy for the application industry to author new applications across a variety of platforms. A good example of this has been the BT Laboratories development of in-flight information, communication and entertainment services, that operate on a number of different on-board operating systems and hardware, provided by each airline. Figure 2 shows the layered nature of an easily portable creation environment.

At the top is the specific application, a shopping catalogue for an imaginary airline. This will have been designed through the story-boarding process described by Smith and Jones [3]. This will be mainly the work of the content and service provider (or their agents) and embody the brand-image, the way they want to show product details, and so on.

This application, as with all applications developed in the environment, will be constrained within certain rules, regarding number of media windows, position and size of buttons, page links, etc, through an ‘application design environment’, essentially a multimedia publishing package. This environment will, in general, be independent of the platform on which the application will run (although specific extensions may be permitted in some circumstances).

This platform independence has been achieved by the identification of a sufficient set of standard procedures that cover the range of activities required to produce the repertoire of applications — procedures such as, ‘fetch page *n*’, ‘link image to window *y*’, etc.

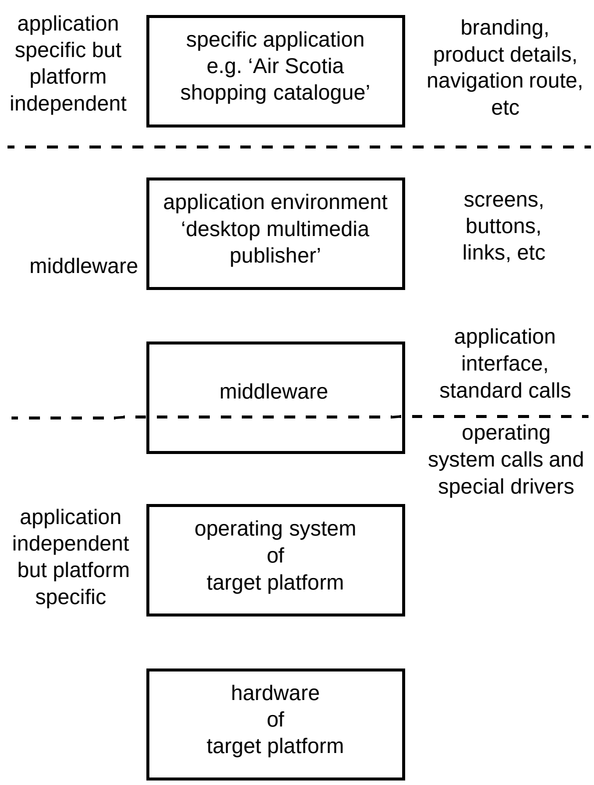

Fig 2 The layered nature of a portable creation environment.

Armed with this application environment and a storyboard, the application developer can create the application without knowing or caring about any details of the platform.

The developer of the application environment, however, in addition to creating the standard, platform-independent application procedures, also has to understand the details of the operating system and hardware of the target platform. Thus the developer must produce a piece of middleware that converts between the generic procedure calls of the application environment (not the application itself) and the realities of the drivers and utilities of the operating system, sometimes even writing directly to the hardware.

Developing this application environment and middleware costs time and money, up-front before any actual application is delivered, and requires foresight and understanding from the customer. However, it is undoubtedly the correct course of action.

4.3 Multimedia structures

Another aspect of standardisation is discussed by Eyles [8]. He considers the handling of multimedia content from a variety of source types generated and presented on a variety of hardware platforms.

He argues that content provision will be a major source of both revenue and costs; this will give rise to a strong desire for reuse and the solution will be to adopt a standard, intermediating format for the transfer of multimedia content. One such format has been addressed by the Multimedia and Hypermedia Information Coding Expert Group (MHEG) in ISO. Data structures held within platforms and applications and coded in private or proprietary format can be recoded and decoded to and from a standard MHEG format, for transferring to the presentation device.

Clearly, the MHEG coding scheme must be flexible enough to support most, if not all, foreseeable multimedia presentations. The coding scheme should support the referencing and/or the encapsulation of objects encoded through existing format such as MPEG video, G.711 audio, etc. It also should allow the composing of complex objects and expressing relationships, such as spatial and temporal position (e.g. subtitles on a moving video), and so on. Adopting this approach, he argues, leads to a better understanding of the distinction between content and application, and removes the reliance on a hardware/ software 'stovepipe' solution.

4.4 Interactive TV

So far, this section has identified a number of system attributes to meet the needs of emotionally satisfying, high-quality content, particularly with a domestic and entertainment slant:

- good media quality,
- cheap receivers,
- asymmetrical distribution,
- store and access,
- standardised construction.

Recently, the label 'interactive TV' has become applied to a range of products that go some way to meeting this list of requirements. They include broadcast satellite, terrestrial and cable TV with some form of backward channel, perhaps PSTN or packet data, and also the emerging technology of video on demand.

4.5 Broadcast television

Dobbie [9] claims that: 'Broadcast television is perhaps the most widely accepted form of multimedia communication today'. The major change from analogue to digital is imminent and, along with this change, comes the opportunity to provide a range of levels of personalisation of service, including backward channels that allow user interaction.

After reviewing current standards, he describes how digital transmission has evolved — the broadcast coding standard has stabilised on MPEG-2, incorporated within a set-top box with separate inputs for terrestrial, satellite and cable delivery. There will also be 2-way connectivity to PSTN or ISDN which will provide the backward channel as well as a means of delivering information services, e.g. Internet, or the downloading of software for games, etc.

Eventually the functionality of the set-top box is likely to be incorporated within the TV set itself, given suitable markets and agreed standards.

Clearly, the two issues are interrelated and the roles of the two key drivers of standards — the European Digital Video Broadcasting Project (DVB), and the Digital Audio-Visual Council (DAVIC) — are discussed.

Membership of these two bodies demonstrates the role of multimedia communications enforcing convergence upon the computing, consumer electronics and telecommunications industries.

4.6 Video on demand

The basic architecture of video on demand is shown in Fig 3. Video material — films, TV programmes, home-shopping material, etc — is stored as compressed digitised

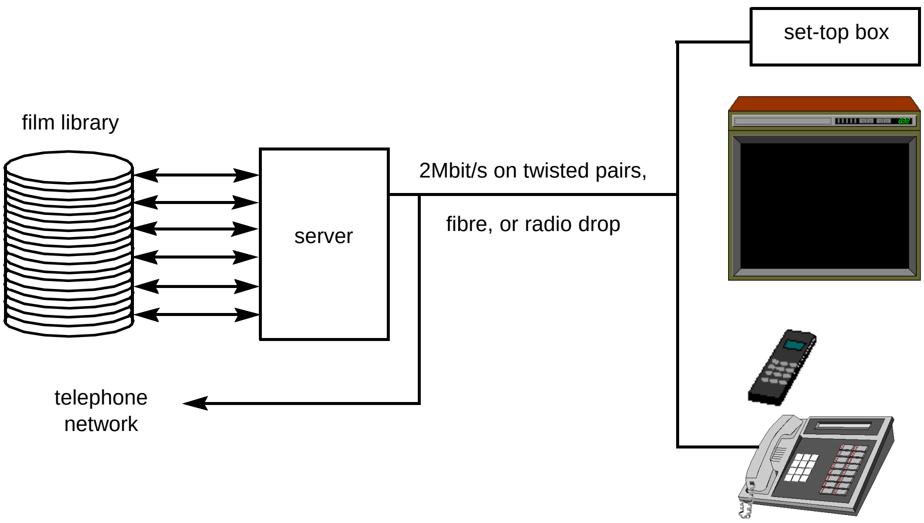

Fig 3 Video on demand basic architecture.

material on a video server. The servers are accessed by set-top boxes (on top of domestic TV sets), on demand of the user, who typically controls the set-top box with a standard TV remote control unit. Requests from the set-top box travel over a relatively slow-speed backward channel to the server which can reply with a continuous stream of data at megabits per second rates and unique to the user — hence the ‘on-demand’ name.

Each of the components of the service sets some significant technology challenges.

- Video servers

  Kerr [10] describes the issues surrounding the design of the video server. Although he also covers some aspects of ‘low bit rate’ servers operating at fractional Mbit/s rates, the main thrust of the paper is on equipment capable of continuously delivering bit rates in excess of 1 Mbit/s per user. Given the large-scale demands for data storage and access — a video library of 2000 hours of material requires 2 Tbytes capacity and 2 Gbit/s access speeds to cover over two thousand simultaneous accesses — it is not surprising that network planning and trade-off between server location and size are major issues.

  Service architecture is also heavily dependent on application; passive viewing places fewer demands than do interactive services such as home shopping.

  The paper reviews the different designs adopted by vendors. Some use special purpose hardware, some employ massively parallel processors while another school of thought believes that multiple small-scale conventional processors offer the low-cost, expandable solution.

- Set-top boxes

  The set-top box architecture and evolution are described by Bissell and Eales [11]. The set-top box is a consumer item whose success is critically dependent on the amount and quality of content available. Consequently, it is essential to standardise the format of the content, as far as possible, across different manufacturers of the same application platform, and also across different applications, e.g. set-top boxes for video on demand and satellite.

  The set-top box used in the BT video on demand trial is an Apple Macintosh LC-series computer. The paper describes the ‘CPU-centric’ design architecture, both software and hardware. It then goes on to predict possible evolution paths for the set-top box, including its ultimate absorption into the TV set.

  The authors believe that the overall design principles will be enduring, with minor variations depending on the split between hardware and software implementations of functional blocks. However, the market will split into at least two classes of box — a basic unit for simple interactive TV capability, with minimal downloading of complex applications, and a high performance unit, based on computer workstation designs, capable of considerable expansion and very high resolution graphics.

- Delivering video on demand to the home

  In terms of data quantity and data rate, video on demand clearly sets some exacting requirements.

  There are a number of transmission techniques under development for delivering the data to the customer’s premises. 2 Mbit/s radio is a possibility, as is fibre, but perhaps the most interesting is the so-called 'asymmetric digital subscriber loop' (ADSL) method of delivery over the existing copper pair. Young, Foster and Cook [12] explain how data rates of several megabits per second over multiple kilometre lengths of unmodified cable have been successfully demonstrated in the field. The 'trick' is simple: by restricting the transmission of high-speed data to one direction, from exchange to customer, there is no problem with near-end cross-talk, which is usually the factor that determines the maximum range in the case of bi-directional transmission.

Even so, it is not possible to achieve acceptable error rates without making use of complex coding and equalising techniques and a number of these exist without a clear winner identified at this time. This is a very active area of investigation and hardware designers can take heart that, beside the digital and signal processing developments, are some very demanding 'real' filter design problems in separating the high frequency VoD signals from the low frequency telephony ones.

4.7 Emotion, decision, commitment — public terminals

The simplest application for interactive TV is probably the selection of movies or television recordings, but it is also easy to see how it could be extended to provide more complex interactive services. One such example is the home shopping scenario, with the triple need to engage, inform and transact with the customer. Interactive television is an obvious vehicle for achieving this, but currently there remain barriers in the way — broadcast and many cable services lack sufficient ability to be personalised to individual requirements; video on demand is still experimental and is currently range limited.

What is required is the media quality of interactive TV with the ubiquity of PSTN.

Totty [13] explores some compromises and synergies, in this direction, in his discussion of public terminals.

The terminals are located in places visited by subsets of the public — customers in a shop, travellers on an aeroplane, in an airport concourse, visitors to a museum, etc. Sometimes they sell goods or try to provide an absorbing experience, sometimes they are more concerned with providing information such as street plans or community services, or providing access to remote expertise.

Terminals can be 'point of information', providing information in predefined order, 'browsers', giving freedom to roam through the database, 'option selectors', where assemblies of goods and services can be created and order tickets produced, and 'full function' kiosks which provide a telecommunications channel to a remote customer-handling point, together with the ability to handle financial transactions and presentation of receipts.

It is the contention of the paper that the natural evolution path is in the direction of 'full functionality'. This is the only way that high-quality images can be combined with a mechanism for committing the customer to a transaction, for example, a purchase.

Traditionally, terminals held their high-quality images locally on mass storage devices such as hard disk or CD-ROM, but this presents some problems, notably the difficulty of issue control if a large number of terminals are involved, and the need to update the database, for example, with product price and availability. This is one area where a communications link can help, even though broadband networks, such as video on demand, are not yet widely available. It can also offer the possibility of a 'pop-up' assistant, whose eye-to-eye involvement with the potential customer may facilitate a sale.

Totty explores a number of options for combining the switched, flexible and controlled network connection with local processing and storage. He also expands on the requirements of kiosks to handle transaction cards, the need for network management and maintenance control, and how they can be integrated into a complete fulfilment operation.

4.8 Utility versus emotion

When the requirements and the solutions so far are compared, a clear pattern emerges — highly absorbing, quality content, with vivid images and sound, can only be delivered at the expense of distance, bi-directional bandwidth, immediacy of generation and flexibility of origin. Figure 4 shows this in a rather idealised form.

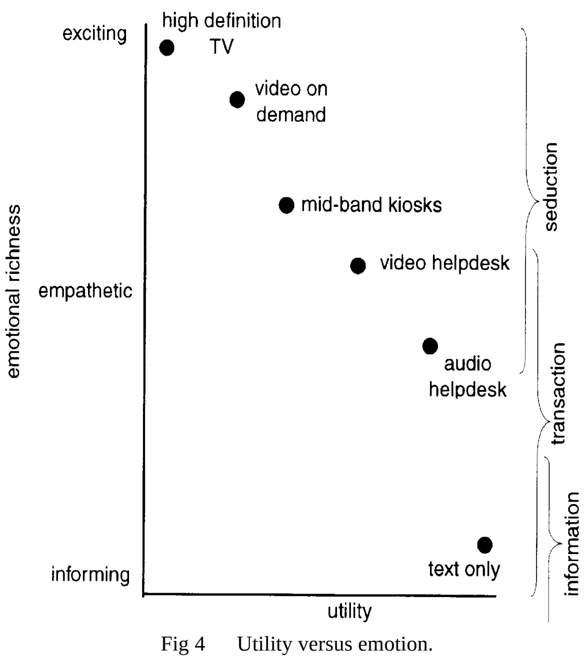

Fig 4 Utility versus emotion.

The vertical axis represents the emotional impact produced by the media. As has been shown, this is highly correlated with media richness and thus bandwidth. (This can be one-way richness, for entertainment services, but both-way bandwidth and latency must be taken into account for empathy-based services, such as conversations.)

The horizontal axis attempts to depict a rather abstract and ill-defined quantity, 'utility', which is to be interpreted as a complicated function of price, usability, ubiquity of access and connectivity between multiple receivers and multiple sources. It is not an exact quantity, but it has proved of practical value in discussions, with most people being consistent in their placing of products and services along the axis, at least when discussing similar applications. It appears in a slightly different form in Totty [13].

Superimposed on Fig 4 are the three zones of seduction, transaction, information. For example, a telephone helpdesk transaction need not be exciting, but there must be a reasonable degree of empathy between the agent and the customer. This would be enhanced by the presence of video telephony, but, of course, current pricing and availability would then reduce the utility.

In general, the goal of multimedia communications is the top, right-hand corner, but different requirements are necessarily positioned at different points from it. As previously described, entertainment services, benchmarked against broadcast TV, have had to sacrifice utility in terms of range (video on demand), connectivity (CD-ROM) and interactivity (satellite and video-tape) for media richness; kiosks have compromised on emotion and utility, but can achieve a top-right shift by introducing a transmission component for video updates and real time interaction.

4.9 Decision environments

Multimedia communications potentially allows us to integrate a number of distributed work-groups into a single, logical office, but any attempt to introduce multimedia communications into the business environment must accept that businesses demand a very high utility from all their processes. Specifically, anyone who has been involved in the introduction of office automation knows that major benefits can only be achieved when a practically ubiquitous service is provided. It is much more important to provide basic e-mail to everyone in the organisation than making available compound, modifiable documents only to a select few. In fact, Metcalfe (one of the pioneers of Ethernet) equates the 'value' of a network (that is, its utility) to the number of users squared [14].

This places severe constraints on the realisation of multimedia in decision environments.

- Teleworking

Perhaps the simplest example of a distributed work-group is the clerical teleworker, who works at home while accessing a remote system over a data link and interacting with colleagues over a videophone. This subject is covered in Gray [15] detailing the well-known 'Inverness Experiment', involving ten BT Directory Assistance operators. One of the major discoveries from this experiment was evidence to support the hypothesis that teleworkers benefit from a rich support environment providing formal and informal contact with colleagues and managers.

The operators were provided with video and voice-only telephony, e-mail and electronic whiteboards. Video was provided over basic rate ISDN. Each of the facilities was used several times a day, the videophone most frequently. Formal psychological testing demonstrated that the environment was indeed highly supportive. The value of videotelephony was graphically illustrated when its temporary withdrawal, for operational reasons, was reflected in a marked reduction in the overall acceptability of the system! A further important conclusion drawn from this and other incidents, is the critical need for reliability and good maintenance procedures.

- Desktop multimedia

The teleworking experiment is an example of a very simple distributed working process, in that the individual operators perform tasks that are not in the main coupled to the tasks of the others and operate as independent clients of a main computer. Many tasks involve a greater degree of co-operative, peer-to-peer working, and many of these would benefit from improved human judgement through the collective use of multimedia.

One such support tool, recently made available by advances in technology, is the desktop multimedia terminal. Midwinter [16] addresses the components necessary for producing desktop equipment and associated control systems, that make multimedia communications a genuine solution to distributed working. Desktop multimedia is much more than just a technology for connecting personal computers to the telephone network to provide simultaneous transmission of video, speech and data. If it is to be genuinely useful, it must provide full telephony functionality, including basic functions such as operation in power-down mode, as well as boss/secretary working.

Also, it must interwork with other units within the company and with other users, world-wide, not just through telecommunications but also office automation. Standards clearly are an issue, and the paper gives a comprehensive account of the various media coding and aggregation standards, and those addressing network control and applications.

4.10 Integration and convergence

As shown earlier, the principal issue that needs to be addressed is the relationship between the multimedia communications service and the total business process it supports. As with any business automation solution, ubiquity of delivery, interoperability with existing services and technologies, scaleability, etc, are more important than high, point functionality.

The prime challenge for decision-supporting multimedia communications is its successful integration into the IT and telecommunications infrastructures that run the business processes.

This implies that our video calls, our picture, video and audio databases should all be addressable within the company- (or even enterprise-) wide IT and communications networks, should be transferable and accessible through these networks in a manner consistent with their non-multimedia data architectures and accessed and controlled by the users' workstations and telephony equipment, via user interfaces compatible with other non-multimedia facilities. In practice, this may mean sacrificing Hollywood standards of images.

Notice that both IT and telecommunications were identified. Multimedia within the business environment is a hybrid between human-to-human interaction and process-driven computing tasks. Its requirements are based on human senses, but it is mediated through computer protocols. Sometimes this can cause problems. The paper by Bunn [17] highlights one such area, the problem of overlong and variable delay in local and wide area networks. These delays cause few problems when computers are communicating with each other, but, on networks with even modest degrees of load, can completely destroy the perceived continuity of moving images or speech. See also Baker and Paradiso [18] for further discussion of desktop multimedia in office environments.

Successful integration of multimedia communications into the business process calls into question the continuing division between telephony and computing.

As a corollary to this tendency of multimedia to force integration between computing and telephony, and the requirement of business to preserve the facilities of both, the development of new services, hybrids of computing and telephony, such as multipoint teleconferencing and multiple terminal interworking, are likely to be seen. Achieving standards for these complex interworkings at the same time as leaving room for competitive edge, across the telecommunications and intensely competitive computing industries, will be crucial.

4.11 Judgement and fun — virtual reality for business

Notwithstanding the dominant need for ubiquitous, utility of service in the business environment, the benefits of rich media presentation should not be dismissed out of hand. Rose and Clarke [19] address the problem that arises with conventional desktop video because the camera is not concentric with the viewing screen. This creates the illusion that the distant party does not want to look you straight in the eye, leading to an adverse reaction to this perceived body language. They discuss a number of solutions to the problem.

At the very least, providing an element of fun within an IT system, leads to significantly greater rates of take-up [20]. The advantages are not necessarily just in their entertainment value; they can contribute to understanding of the content and to navigation through it. In fact, because images can be used to represent logical, rather than physical worlds, things can be placed close together that share a close relationship rather than those that are close simply by an accident of geography.

Rogers [21] describes a virtual reality 'technical centre', which provides a common support service for documents and other information, to a number of factories around the world. Through its authorship, readership, etc, as well as its main text, a document's structure gives valuable clues to the way the work is structured. Representing the features of documents by geometric co-ordinates, the project displays them as clusters of similarity, showing for example, the common user-groups.

Representations of the human interaction in the virtual business can be extended to creating virtual offices which locate people on the basis of the project structure rather than physical constraints. Anyone working on the project can enter the space and see who else is available.

Also in the paper is a description of a virtual fashion enterprise, where text-based information, images of cloth, yarns, etc, are integrated into a process description of the end-to-end business, even including a virtual fashion show with a virtual model on a virtual catwalk, modelling the garment.

5. Communications — options and issues

The role of ATM for transporting multimedia has been addressed elsewhere [22] and several of the papers in this issue also describe transmission requirements and solutions. Therefore, this section will only attempt to summarise the options and draw attention to areas not otherwise covered.

5.1 Delivering entertainment services

The major transmission breakthrough that may make video on demand realisable in the near-term, is ADSL over existing copper cables into the home [12]. However, used alone, this would require servers to be installed within a few kilometres of every customer, which would be expensive and exacerbates the problem of loading them with content, which is a significant transmission capacity task also. Thus, to make ADSL useful, it needs to be incorporated with other transmission technologies. One possible configuration is shown in Fig 5.

ADSL is still used for the last few kilometres, but is connected to an SDH or ATM demultiplexer. This multiplexer is in turn connected to an SDH or ATM switch, which is one of many on a wide-area SDH or ATM ring which also includes the media servers containing the content material to be accessed. As an alternative to ADSL for local delivery, fibre, radio or coaxial cable can obviously be fitted into this structure.

Under wide-scale deployment, the structure may be more complex, with central servers containing information for very wide area distribution, ‘metropolitan servers’ covering, as the name implies, major conurbations, and local servers, providing frequently accessed or local information. See Furht et al [23] for a detailed description, albeit predominantly from a USA perspective, of how telecommunications and cable networks will evolve to this model.

There still remains a lot to be understood in the dimensioning of multimedia servers and their associated networks.

Bi-directional, high-speed transmission to the home is also emerging in the form of ‘HDSL’. If this is successful, it may come about that interactive TV components become the enablers of mass videophone.

Another area requiring further development is in the best choice of distribution medium for multimedia services within the home.

There is no clear leader in the home distribution market, either in terms of transmission medium (coax, cable pair, radio), protocol (ADSL, ATM, LAN, e.g. token ring), or vendor paradigm (computer, telecommunications or consumer electronics).

To a large extent, this will be determined by the winning structure for the home multimedia terminal — interactive TV set-top box, or home computer, in particular.

5.2 Transmission techniques for business multimedia and kiosks

Unlike entertainment networks, the requirements for distributed working depend less on very wideband (though predominantly uni-directional) provision and more on bi-directional, low-delay, ubiquity of service. To a lesser

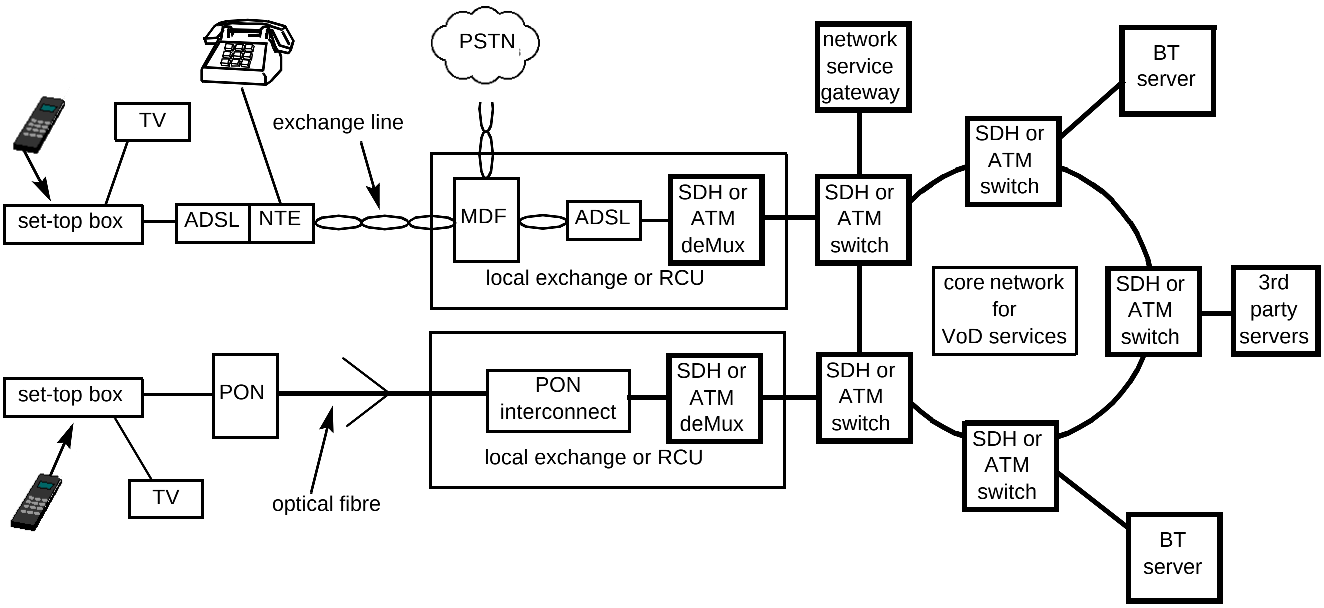

Fig 5 Video on demand network architecture.

extent, this is also true of full-function kiosks. In the short term at least, this is a significant opportunity for ISDN, certainly at basic rate and perhaps with aggregated channels of six or even more.

However, on its own, this is not much more than a videoconferencing technique, or, at most, a solution to the interconnection of executive desktop terminals connected to direct exchange lines. As explained above, multimedia communications also needs to be integrated into the IT infrastructure. This needs a rethink to LAN architectures, in order to provide guaranteed bandwidth and low delay. Given that ATM may be the natural evolution path for distributed working multimedia over the wide area, it will be interesting to see how far it penetrates intra-building standards.

5.3 Multimedia on the move

The subject of multimedia communications techniques would not be complete without mentioning two successful applications which required wireless communication on the move. There almost seems something perverse in the way BT has tackled the issue of providing multimedia content-based services — the first offerings have been (almost literally) launched from aircraft and from yachts at sea!

The in-flight system for information, shopping and entertainment services has already been mentioned. This consists of a local multimedia platform located on the aircraft, with a (non-multimedia) air-to-ground (direct or via satellite) link to carry data traffic, such as the purchase orders for goods or the despatching of facsimile messages. Use of the data channel, which operates over the Skyphone/Jetphone service [24], converts the in-flight service from a browser to a full-function terminal.

Compared with the stability of an airborne platform, bringing back pictures from yachts at sea presents an even greater challenge. Woolf and Tilson [25] describe how moving pictures and sound, to a quality acceptable for TV broadcasts, were composed on-board and then transmitted at a low bit rate (64 kbit/s) reliably over a satellite link, despite the extremely adverse transmitting conditions, using a novel error-correction system.

6. Convergence and conclusions

So far it has been shown that the difficulty in defining multimedia communications has come about by the varying and significantly different human requirements being met by tactical selections of technologies chosen to meet only the key human requirements. Content creation has been shown to bring together the needs and aspirations of a number of different and differing parties. Also clear is the platform issue — TV/set-top box, or personal computer — and the different transmission parameters of telecommunications and computing networks.

Resolving these issues and achieving harmonisation at the technology level, while leaving room for the development of a wide choice of products, is the domain of the so-called ‘convergence’ debate.

When, and not before, they are resolved, then and only then, will the ‘Information Superhighway’ really come into being, providing the ubiquitous delivery of entertainment, information and instantaneous, bi-directional, media-rich communications. At that time, we will at last easily understand what is meant by ‘multimedia communications’.

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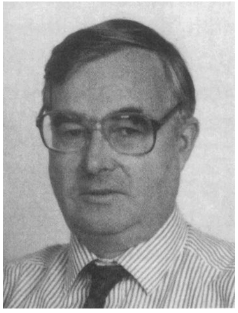

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